

Puppet Mastery Part-7, Practical 16 (Flagship Practical-5) - Enterprise Load Balancing, High Availability, and Horizontal Scaling with Puppet and HAProxy

Note: Since this is a dynamic load balancing practical have merged all the 4 instances in a way similar to how exactly the practical was done

```
ubuntu@puppet-master: ~  
ubuntu@puppet-master:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.10.104)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-master  
Info: Applying configuration version '1786437078'  
Notice: /Stage[main]/Loadbalancer::Config/File[/etc/haproxy/haproxy.cfg]/content  
:  
--- /etc/haproxy/haproxy.cfg      2025-12-03 16:00:30.000000000 +0000  
+++ /tmp/puppet-file20260811-7680-1y010i      2026-08-11 08:31:19.415213569 +0000  
@@ -1,34 +1,23 @@  
global  
-     log /dev/log      local0  
-     log /dev/log      local1 notice  
-     chroot /var/lib/haproxy  
-     stats socket /run/haproxy/admin.sock mode 660 level admin expose-fd list  
eners  
-     stats timeout 30s  
-     user haproxy  
-     group haproxy  
-     daemon  
+ log /dev/log local0
```

```

ubuntu@puppet-master: ~
- stats socket /run/haproxy/admin.sock mode 660 level admin expose-fd list
eners
- stats timeout 30s
- user haproxy
- group haproxy
- daemon
+ log /dev/log local0
+ log /dev/log local1 notice
+ daemon

- # Default SSL material locations
- ca-base /etc/ssl/certs
- crt-base /etc/ssl/private
+defaults
+ log global
+ mode http
+ option httplog
+ option dontlognull
+ timeout connect 5s
+ timeout client 30s
+ timeout server 30s

- # See: https://ssl-config.mozilla.org/#server=haproxy&server-version=2.0
.3&config=intermediate
- ssl-default-bind-ciphers ECDHE-ECDSA-AES128-GCM-SHA256:ECDHE-RSA-AES128

```

```

ubuntu@puppet-master: ~
+ default_backend web_servers

-defaults
- log global
- mode http
- option httplog
- option dontlognull
- timeout connect 5000
- timeout client 50000
- timeout server 50000
- errorfile 400 /etc/haproxy/errors/400.http
- errorfile 403 /etc/haproxy/errors/403.http
- errorfile 408 /etc/haproxy/errors/408.http
- errorfile 500 /etc/haproxy/errors/500.http
- errorfile 502 /etc/haproxy/errors/502.http
- errorfile 503 /etc/haproxy/errors/503.http
- errorfile 504 /etc/haproxy/errors/504.http
+backend web_servers
+ balance roundrobin
+ option httpchk GET /
+ server web01 172.31.0.173:80 check
+ server web02 172.31.5.59:80 check

Notice: /Stage[main]/Loadbalancer::Config/File[/etc/haproxy/haproxy.cfg]/content
: content changed '{sha256}90977f5a5f97231a1b03e6e9510e7bc98f5e42c8f08cf25a4eb7c

```

```
ubuntu@puppet-master: ~
-       errorfile 503 /etc/haproxy/errors/503.http
-       errorfile 504 /etc/haproxy/errors/504.http
+backend web_servers
+  balance roundrobin
+  option httpchk GET /
+  server web01 172.31.0.173:80 check
+  server web02 172.31.5.59:80 check

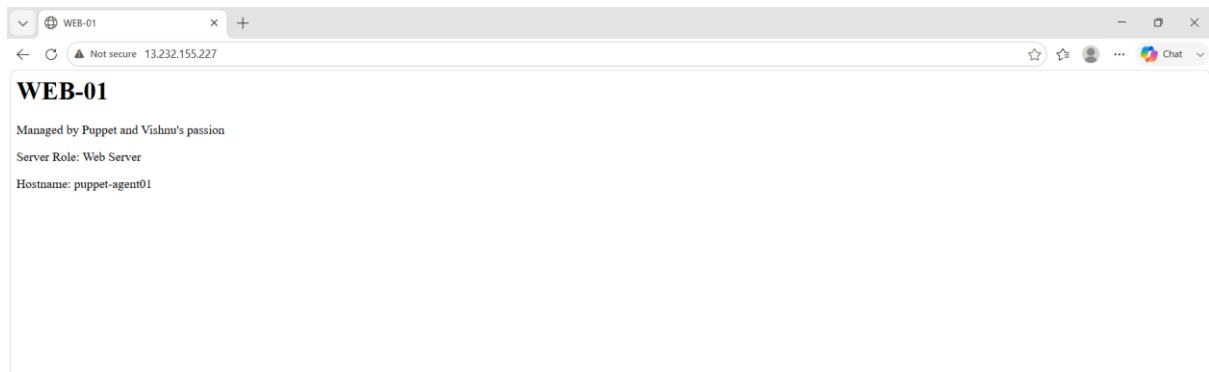
Notice: /Stage[main]/Loadbalancer::Config/File[/etc/haproxy/haproxy.cfg]/content
: content changed '{sha256}90977f5a5f97231a1b03e6e9510e7bc98f5e42c8f08cf25a4eb7c
8fae4d55079' to '{sha256}71b62faaafa008f236418a9d0ef7eab3ac624deb95c9225bb5b19b7
2a106cd63'
Info: /Stage[main]/Loadbalancer::Config/File[/etc/haproxy/haproxy.cfg]: Scheduling refresh of Service[haproxy]
Notice: /Stage[main]/Loadbalancer::Service/Service[haproxy]: Triggered 'refresh'
from 1 event
Info: Creating state file /opt/puppetlabs/puppet/cache/state/state.yaml
Notice: Applied catalog in 0.55 seconds
ubuntu@puppet-master:~$ sudo systemctl status haproxy
● haproxy.service - HAProxy Load Balancer
   Loaded: loaded (/lib/systemd/system/haproxy.service; enabled; vendor prese
   Active: active (running) since Tue 2026-08-11 08:31:19 UTC; 48s ago
     Docs: man:haproxy(1)
           file:/usr/share/doc/haproxy/configuration.txt.gz
   Process: 7749 ExecStartPre=/usr/sbin/haproxy -Ws -f $CONFIG -c -q $EXTRAOPT
```

```
ubuntu@puppet-master: ~
Memory: 69.4M
CPU: 118ms
CGroup: /system.slice/haproxy.service
└─7753 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/h
└─7755 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -p /run/h

Aug 11 08:31:19 puppet-master systemd[1]: Starting HAProxy Load Balancer...
Aug 11 08:31:19 puppet-master haproxy[7753]: [NOTICE] (7753) : New worker #1
Aug 11 08:31:19 puppet-master systemd[1]: Started HAProxy Load Balancer.
ubuntu@puppet-master:~$ sudo ss -tlnp | grep :80
LISTEN 0      4096          0.0.0.0:80          0.0.0.0:*        users:(("haproxy",pid=
7755,fd=6))
ubuntu@puppet-master:~$ for i in {1..10}; do curl -s http://localhost | grep '<h1>'; done
<h1>WEB-01</h1>
<h1>WEB-02</h1>
<h1>WEB-01</h1>
<h1>WEB-02</h1>
<h1>WEB-01</h1>
<h1>WEB-02</h1>
<h1>WEB-01</h1>
<h1>WEB-02</h1>
<h1>WEB-01</h1>
<h1>WEB-02</h1>
ubuntu@puppet-master:~$ |
```

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.10.104)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1786433280'  
Notice: /Stage[main]/Dynamicinfra::Config/File[/var/www/html/index.html]/content  
:  
--- /var/www/html/index.html 2026-08-10 18:11:52.223699814 +0000  
+++ /tmp/puppet-file20260811-1688-fvew2 2026-08-11 07:28:02.862315075 +0000  
@@ -1,363 +1,11 @@  
-<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">  
-<html xmlns="http://www.w3.org/1999/xhtml">  
-  <!--  
-    Modified from the Debian original for Ubuntu  
-    Last updated: 2022-03-22  
-    See: https://launchpad.net/bugs/1966004  
-  -->  
-  <head>  
-    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />  
-    <title>Apache2 Ubuntu Default Page: It works</title>
```

```
ubuntu@puppet-agent01: ~  
+<p>Server Role: Web Server</p>  
+<p>Hostname: puppet-agent01</p>  
+</body>  
+</html>  
  
Notice: /Stage[main]/Dynamicinfra::Config/File[/var/www/html/index.html]/content  
: content changed '{sha256}6faef4d5d777fdcaa653766b0ac8b9ed32d0fd87f7dcd79f02ff5  
24dd1b0eb69' to '{sha256}739003293d921782ca001609b8138eb5d149f71d2a90d34a4eb908f  
9603c703b'  
Notice: Applied catalog in 1.49 seconds  
ubuntu@puppet-agent01:~$ curl http://localhost  
<html>  
<head>  
<title>WEB-01</title>  
</head>  
<body>  
<h1>WEB-01</h1>  
<p>Managed by Puppet and Vishnu's passion</p>  
<p>Server Role: Web Server</p>  
<p>Hostname: puppet-agent01</p>  
</body>  
</html>  
ubuntu@puppet-agent01:~$ hostname -i  
172.31.0.173 fe80::85e:5dff:fe97:ddb
```



```
ubuntu@puppet-agent01:~$ sudo systemctl stop apache2
ubuntu@puppet-agent01:~$ sudo systemctl is-active apache2
inactive
ubuntu@puppet-agent01:~$ |
```

```
ubuntu@puppet-master: ~
ubuntu@puppet-master:~$ for i in {1..10}; do curl -s http://localhost | grep '<h1>'; done
<h1>WEB-01</h1>
<h1>WEB-02</h1>
<h1>WEB-01</h1>
<h1>WEB-02</h1>
<h1>WEB-01</h1>
<h1>WEB-02</h1>
<h1>WEB-01</h1>
<h1>WEB-02</h1>
<h1>WEB-01</h1>
<h1>WEB-02</h1>
ubuntu@puppet-master:~$ for i in {1..10}; do curl -s http://localhost | grep '<h1>'; done
<h1>WEB-02</h1>
<h1>WEB-02</h1>
<h1>WEB-02</h1>
<h1>WEB-02</h1>
<h1>WEB-02</h1>
<h1>WEB-02</h1>
<h1>WEB-02</h1>
<h1>WEB-02</h1>
<h1>WEB-02</h1>
ubuntu@puppet-master:~$ |
```

Note: Load balancer detected that web-01 health check fails so traffic goes only to WEB-02

```
ubuntu@puppet-agent01: ~  
<p>Hostname: puppet-agent01</p>  
</body>  
</html>  
ubuntu@puppet-agent01:~$ hostname -i  
172.31.0.173 fe80::85e:5dff:fe97:ddb  
ubuntu@puppet-agent01:~$ sudo systemctl stop apache2  
ubuntu@puppet-agent01:~$ sudo systemctl is-active apache2  
inactive  
ubuntu@puppet-agent01:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.10.104)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1786437904'  
Notice: /Stage[main]/Dynamicinfra::Service/Service[apache2]/ensure: ensure chang  
ed 'stopped' to 'running' (corrective)  
Info: /Stage[main]/Dynamicinfra::Service/Service[apache2]: Unscheduling refresh  
on Service[apache2]  
Notice: Applied catalog in 0.99 seconds  
ubuntu@puppet-agent01:~$ sudo systemctl is-active apache2  
active  
ubuntu@puppet-agent01:~$
```

```
ubuntu@puppet-master:~$ for i in {1..10}; do curl -s http://localhost | grep '<h  
1>'; done  
<h1>WEB-02</h1>  
<h1>WEB-01</h1>  
<h1>WEB-02</h1>  
<h1>WEB-01</h1>  
<h1>WEB-02</h1>  
<h1>WEB-01</h1>  
<h1>WEB-02</h1>  
<h1>WEB-01</h1>  
<h1>WEB-02</h1>  
<h1>WEB-01</h1>  
ubuntu@puppet-master:~$ |
```

Note: After desired state was revived through catalog compilation and apache2 was made active, load balancing again starts distributing evenly

```
ubuntu@puppet-agent03: ~  
ubuntu@puppet-agent03:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.10.104)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent03  
Info: Applying configuration version '1786451350'  
Notice: /Stage[main]/Dynamicinfra::Install/Package[tree]/ensure: created  
Notice: /Stage[main]/Dynamicinfra::Install/Package[apache2]/ensure: created  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company]/ensure: created  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]/ensure: defined content as '{sha256}7bd79343d1410394c76293c1d1c70a611cbfa713bfea8c25b23948cceb002ab5'  
Info: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]: Scheduling refresh of Service[apache2]  
Notice: /Stage[main]/Dynamicinfra::Config/File[/var/www/html/index.html]/content:  
--- /var/www/html/index.html      2026-08-11 12:29:26.613767857 +0000  
+++ /tmp/puppet-file20260811-1022-wb6mxu      2026-08-11 12:29:33.587808987 +0000  
@@ -1,363 +1,11 @@  
-<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">  
+<html>  
+<head>  
+<title>WEB-03</title>  
+</head>  
+<body>  
+<h1>WEB-03</h1>  
+<p>Managed by Puppet and Vishnu's passion</p>  
+<p>Server Role: Web Server</p>  
+<p>Hostname: puppet-agent03</p>  
+</body>  
+</html>
```

```
ubuntu@puppet-agent03: ~  
+<html>  
+<head>  
+<title>WEB-03</title>  
+</head>  
+<body>  
+<h1>WEB-03</h1>  
+<p>Managed by Puppet and Vishnu's passion</p>  
+<p>Server Role: Web Server</p>  
+<p>Hostname: puppet-agent03</p>  
+</body>  
+</html>  
Notice: /Stage[main]/Dynamicinfra::Config/File[/var/www/html/index.html]/content: content changed '{sha256}6faef4d5d777fdcaa653766b0ac8b9ed32d0fd87f7dcd79f02ff524dd1b0eb69' to '{sha256}7f7421814dd4278070f9cc121b4fce1ffdd7bca31d070cb06954105118749964'  
Notice: /Stage[main]/Dynamicinfra::Service/Service[apache2]: Triggered 'refresh' from 1 event  
Notice: /Stage[main]/Dynamicinfra::Report/File[/opt/company/deployment-report.txt]/ensure: defined content as '{sha256}f92a371da788e0e51af22f548239142afe55a1f45132a0d85e2f8d191f8e04cd'  
Info: Creating state file /opt/puppetlabs/puppet/cache/state/state.yaml  
Notice: Applied catalog in 17.73 seconds  
ubuntu@puppet-agent03:~$
```



```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/dat...
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/data/nodes$ su
do /opt/puppetlabs/bin/puppet agent -t
Info: Using environment 'production'
Info: Retrieving pluginfacts
Info: Retrieving plugin
Notice: Requesting catalog from puppet-master:8140 (172.31.10.104)
Notice: Catalog compiled by puppet-master
Info: Caching catalog for puppet-master
Info: Applying configuration version '1786453356'
Notice: /Stage[main]/Loadbalancer::Config/File[/etc/haproxy/haproxy.cfg]/content
:
--- /etc/haproxy/haproxy.cfg      2026-08-11 12:56:27.825271104 +0000
+++ /tmp/puppet-file20260811-1792-11i715      2026-08-11 13:02:38.155902439 +0
000
@@ -19,6 +19,6 @@
  backend web_servers
    balance roundrobin
    option httpchk GET /
-   server 172.31.0.173 172.31.0.173:80 check
-   server 172.31.5.59 172.31.5.59:80 check
-   server 172.31.12.70 172.31.12.70:80 check
+   server web01 172.31.0.173:80 check
+   server web02 172.31.5.59:80 check
+   server web03 172.31.12.70:80 check

```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/dat...
--- /etc/haproxy/haproxy.cfg      2026-08-11 12:56:27.825271104 +0000
+++ /tmp/puppet-file20260811-1792-11i715      2026-08-11 13:02:38.155902439 +0
000
@@ -19,6 +19,6 @@
  backend web_servers
    balance roundrobin
    option httpchk GET /
-   server 172.31.0.173 172.31.0.173:80 check
-   server 172.31.5.59 172.31.5.59:80 check
-   server 172.31.12.70 172.31.12.70:80 check
+   server web01 172.31.0.173:80 check
+   server web02 172.31.5.59:80 check
+   server web03 172.31.12.70:80 check

Notice: /Stage[main]/Loadbalancer::Config/File[/etc/haproxy/haproxy.cfg]/content
: content changed '{sha256}7a76cabad5afe14e0275bf47b4c559088bf3a9eff541c07315a50
bc45f655336' to '{sha256}171f5cf75bac76f68cf504d5b7c69a66142fbd6f81577b1216a887e
08d7416eb'
Info: /Stage[main]/Loadbalancer::Config/File[/etc/haproxy/haproxy.cfg]: Scheduling refresh of Service[haproxy]
Notice: /Stage[main]/Loadbalancer::Service/Service[haproxy]: Triggered 'refresh'
from 1 event
Notice: Applied catalog in 0.54 seconds
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/data/nodes$

```



```
ubuntu@puppet-master: ~  
Aug 11 12:28:34 puppet-master puppetserver[1192]: 12:28:34,676 |-INFO in ch.qos>  
Aug 11 12:28:34 puppet-master puppetserver[1192]: 12:28:34,676 |-INFO in ch.qos>  
Aug 11 12:28:34 puppet-master puppetserver[1192]: 12:28:34,676 |-INFO in ch.qos>  
Aug 11 12:28:34 puppet-master puppetserver[1192]: 12:28:34,676 |-INFO in ch.qos>  
Aug 11 12:28:34 puppet-master puppetserver[1192]: 12:28:34,677 |-INFO in ch.qos>  
Aug 11 12:28:35 puppet-master systemd[1]: Started puppetserver Service.  
ubuntu@puppet-master:~$ for i in {1..15}; do curl -s http://localhost | grep '<h1>'; done  
<h1>WEB-01</h1>  
<h1>WEB-02</h1>  
<h1>WEB-03</h1>  
<h1>WEB-01</h1>  
<h1>WEB-02</h1>  
<h1>WEB-03</h1>  
<h1>WEB-01</h1>  
<h1>WEB-02</h1>  
<h1>WEB-03</h1>  
<h1>WEB-01</h1>  
<h1>WEB-02</h1>  
<h1>WEB-03</h1>  
<h1>WEB-01</h1>  
<h1>WEB-02</h1>  
<h1>WEB-03</h1>  
ubuntu@puppet-master:~$ |
```

Successful Three-Node Horizontal Load Balancing Using HAProxy

All key files of this beautiful practical:

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments  
./development  
./development/manifests  
./development/manifests/site.pp  
./development/modules  
./development/modules/dynamicinfra  
./development/modules/dynamicinfra/templates  
./development/modules/dynamicinfra/manifests  
./development/modules/dynamicinfra/functions  
./production  
./production/data  
./production/data/nodes  
./production/data/nodes/puppet-agent01.yaml.save  
./production/data/nodes/puppet-agent01.yaml  
./production/data/nodes/puppet-agent03.yaml  
./production/data/nodes/puppet-master.yaml  
./production/data/nodes/puppet-agent02.yaml  
./production/data/production.yaml  
./production/data/production.yaml.save  
./production/hiera.yaml  
./production/manifests  
./production/manifests/site.pp  
./production/modules  
./production/modules/dynamicinfra  
./production/modules/dynamicinfra/templates
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments
/production/modules/dynamicinfra
/production/modules/dynamicinfra/templates
/production/modules/dynamicinfra/templates/system-info.epp
/production/modules/dynamicinfra/templates/deployment-report.epp
/production/modules/dynamicinfra/templates/backend.epp
/production/modules/dynamicinfra/manifests
/production/modules/dynamicinfra/manifests/install.pp
/production/modules/dynamicinfra/manifests/config.pp
/production/modules/dynamicinfra/manifests/service.pp
/production/modules/dynamicinfra/manifests/init.pp
/production/modules/dynamicinfra/manifests/report.pp
/production/modules/dynamicinfra/functions
/production/modules/dynamicinfra/functions/normalize_role.pp
/production/modules/dynamicinfra/functions/deployment_identity.pp
/production/modules/dynamicinfra/functions/environment_label.pp
/production/modules/dynamicinfra/functions/package_manager.pp
/production/modules/loadbalancer
/production/modules/loadbalancer/templates
/production/modules/loadbalancer/templates/haproxy.cfg.epp
/production/modules/loadbalancer/manifests
/production/modules/loadbalancer/manifests/install.pp
/production/modules/loadbalancer/manifests/config.pp
/production/modules/loadbalancer/manifests/service.pp
/production/modules/loadbalancer/manifests/init.pp
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data$ cd nodes
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data/nodes$ ls
puppet-agent01.yaml      puppet-agent02.yaml  puppet-master.yaml
puppet-agent01.yaml.save puppet-agent03.yaml
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data/nodes$ cat puppet-agent01.yaml
dynamicinfra::server_role: "Web Server"
dynamicinfra::node_label: "WEB-01"
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data/nodes$ cat puppet-agent02.yaml
dynamicinfra::server_role: "Web Server"
dynamicinfra::node_label: "WEB-02"
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data/nodes$ cat puppet-agent03.yaml
dynamicinfra::server_role: "Web Server"
dynamicinfra::node_label: "WEB-03"
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data/nodes$ cat puppet-master.yaml
loadbalancer::backends:
  web01: "172.31.0.173"
  web02: "172.31.5.59"
  web03: "172.31.12.70"
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/data/nodes$ cd ..
```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
loadbalancer::backends:
  web01: "172.31.0.173"
  web02: "172.31.5.59"
  web03: "172.31.12.70"
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/data/nodes$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/data$ cat production.yaml
dynamicinfra::environment_name: "Production"
dynamicinfra::output_directory: "/opt/company"
dynamicinfra::packages:
  - tree
  - curl
  - htop
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/data$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ cd modules
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules$ cd loadbalancer
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadbalancer$ cd manifests
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadbalancer/manifests$ cat init.pp
class loadbalancer(Hash[String,String] $backends){
  class{'loadbalancer::install':}

```

```

class{'loadbalancer::config':
  backends=>$backends
}
class{'loadbalancer::service':}

Class['loadbalancer::install']->Class['loadbalancer::config']->
Class['loadbalancer::service']
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadbalancer/manifests$ cat config.pp
class loadbalancer::config(
  Hash[String,String] $backends
){
  file{'/etc/haproxy/haproxy.cfg':
    ensure=>file,
    owner=>'root',
    group=>'root',
    mode=>'0644',
    content=>epp(
      'loadbalancer/haproxy.cfg.epp',
      {
        'backends'=>$backends
      }
    ),

```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
Hash[String,String] $backends
){
file{'/etc/haproxy/haproxy.cfg':
  ensure=>file,
  owner=>'root',
  group=>'root',
  mode=>'0644',
  content=>epp(
    'loadbalancer/haproxy.cfg.epp',
    {
      'backends'=>$backends
    }
  ),
  notify=>Service['haproxy']
}
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba
lancer/manifests$ cat service.pp
class loadbalancer::service{
  service{'haproxy':
    ensure=>running,
    enable=>true
  }
}
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba
lancer/templates$ cat haproxy.cfg.epp
global
  log /dev/log local0
  log /dev/log local1 notice
  daemon

defaults
  log global
  mode http
  option httplog
  option dontlognull
  timeout connect 5s
  timeout client 30s
  timeout server 30s

frontend http_front
  bind *:80
  default_backend web_servers

backend web_servers
  balance roundrobin
  option httpchk GET /
<% $backends.each |$name,$ip| { -%>
```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
backend web_servers
  balance roundrobin
  option httpchk GET /
<% $backends.each |$name,$ip| { -%>
  server <%= $name %> <%= $ip %>:80 check
<% } -%>
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba
lancer/templates$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/loadba
lancer$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules$ cd dy
namicinfra
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
cinfra$ cd manifests
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
cinfra/manifests$ cat init.pp
class dynamicinfra(
  String $server_role,
  String $environment_name,
  String $output_directory,
  Array[String] $packages,
  String $node_label
){

```

```

  packages=>$packages
}

class{'dynamicinfra::config':
  server_role=>$server_role,
  environment_name=>$environment_name,
  output_directory=>$output_directory,
  node_label=>$node_label
}

class{'dynamicinfra::service': }

class{'dynamicinfra::report':
  server_role=>$server_role,
  environment_name=>$environment_name,
  output_directory=>$output_directory
}

Class['dynamicinfra::install']->Class['dynamicinfra::config']
->Class['dynamicinfra::service']->Class['dynamicinfra::report']
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
cinfra/manifests$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami

```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/dynamicinfra/templates$ cat backend.epp
<html>
<head>
<title><%= $node_label %></title>
</head>
<body>
<h1><%= $node_label %></h1>
<p>Managed by Puppet and Vishnu's passion</p>
<p>Server Role: <%= $server_role %></p>
<p>Hostname: <%= $hostname %></p>
</body>
</html>
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/dynamicinfra/templates$ cd ..
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/dynamicinfra$ sudo nano manifests/config.pp
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/dynamicinfra$ cat manifests/config.pp
class dynamicinfra::config(
  String $server_role,
  String $environment_name,
  String $output_directory,
  String $node_label

```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/dynamicinfra$ cat manifests/config.pp
$os_family=$facts['os']['family']
$package_manager=dynamicinfra::package_manager($os_family)
$os_name=$facts['os']['name']
$hostname=$facts['networking']['hostname']
$ip=$facts['networking']['ip']

$service_name=$facts['os']['family']?{
  'debian'=>'apache2',
  'redhat'=>'httpd',
  'default'=>'apache2'
}

file{$output_directory:
  ensure=>directory,
  owner=>'root',
  group=>'root',
  mode=>'0755'
}

file{"${output_directory}/system-info.txt":
  ensure=>file,
  owner=>'root',
  group=>'root',
  mode=>'0644',

```



```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
content=>epp(
  'dynamicinfra/system-info.epp',
  {
    'server_role'=>$server_role,
    'normalized_role'=>$normalized_role,
    'environment_label'=>$environment_label,
    'deployment_identity'=>$deployment_identity,
    'environment_name'=>$environment_name,
    'os_family'=>$os_family,
    'os_name'=>$os_name,
    'hostname'=>$hostname,
    'ip'=>$ip,
    'package_manager'=>$package_manager
  }
),
notify=>Service[$service_name]
}

file{'/var/www/html/index.html':
  ensure=>file,
  owner=>'root',
  group=>'root',
  mode=>'0644',
  content=>epp(
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
    }
  ),
  notify=>Service[$service_name]
}

file{'/var/www/html/index.html':
  ensure=>file,
  owner=>'root',
  group=>'root',
  mode=>'0644',
  content=>epp(
    'dynamicinfra/backend.epp',
    {
      'node_label'=>$node_label,
      'server_role'=>$server_role,
      'hostname'=>$hostname
    }
  )
}
}

ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dynami
cinfra$
```


WEB-01

15.207.19.158

Chat

WEB-01

Managed by Puppet and Vishnu's passion

Server Role: Web Server

Hostname: puppet-agent01

WEB-02

13.235.67.199

Chat

WEB-02

Managed by Puppet and Vishnu's passion

Server Role: Web Server

Hostname: puppet-agent02

WEB-03

13.201.225.105

Chat

WEB-03

Managed by Puppet and Vishnu's passion

Server Role: Web Server

Hostname: puppet-agent03

Instances (4/4) Info

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

< 1 >

☒	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	F
☒	Puppet-agent01	i-0f36f812820591f65	Running	t3.micro	Initializing	ap-south-1b	ec2-15-252-71-154.ap-...	1
☒	Puppet-agent02	i-0b732d87e6ad15337	Running	t3.micro	Initializing	ap-south-1b	ec2-13-203-195-157.ap-...	1
☒	advanced-pup...	i-02f33be2d0e6ed7f1	Running	t3.small	Initializing	ap-south-1b	ec2-52-66-189-152.ap-...	5
☒	puppet-agent03	i-01d9b01178d7926d0	Running	t3.micro	Initializing	ap-south-1b	ec2-3-111-39-187.ap-s...	3

